Fast Nearest Subspace Search via Random Angular Hashing

Yi Xu, Xianglong Liu, Binshuai Wang, Renshuai Tao, Ke Xia, and Xianbin Cao

***Abstract*—Subspaces frequently offer powerful representation in many tasks including recognition, retrieval, and optimization. In these tasks, the nearest subspaces (i.e., subspace-to-subspace search) often inevitably arise. Several studies in the literature have attempted to address this hard problem using techniques such as locality-sensitive hashing. Unfortunately, these subspace hashing methods are severely affected by poor scaling, with consequently high computational cost or unsatisfying accuracy, when the subspaces originally distribute with arbitrary dimensions. Accordingly, in this paper, we propose random angular hashing, a new and efficient type of locality-sensitive hashing, for linear subspaces of arbitrary dimension. The method we proposed preserves the angular distances among subspaces by randomly projecting their orthonormal basis and then encoding them with binary codes, meanwhile not only achieving fast computation but also maintaining a powerful collision probability. Moreover, its flexibility to easily get a balance between efficiency and accuracy in terms of performance. The extensive experimental results on tasks of face recognition, video de-duplication, and gesture recognition demonstrate that the proposed approach performs better than the state-of-the-art methods heavily, in terms of both accuracy and efficiency (up to 16× speedup).**

***Index Terms*—Large-scale search, nearest subspace search, locality-sensitive hash, linear subspace, subspace hashing.**

I. Introduction

Subspaces naturally have a strong ability to describe the intrinsic structures of data. As a consequence, they usually provide convenient means to represent data in the fields of pattern recognition, computer vision, and statistical learning. There are many typical applications such as face recognition [1], [2], person re-identification [3], video retrieval [4], 2D/3D matching [5], [6], speech emotion recognition [7] and cross-modal retrieval [8]. Among these applications, finding the nearest subspaces (i.e., subspace-to-subspace search) is a necessary problem that inevitably arises; accordingly, when the amount of data involved explodes, seeking the large-scale subspace database out with high ambient dimensions will drastically elevate storage and computational costs [9], [10].

Locality-sensitive hashing (LSH) well balances the computational efficiency and search performance using short hash codes, being regarded as one of the most powerful approaches for fast nearest neighbor search over a large-scale database. [11] introduced the linear projection paradigm to produce the hash codes for the specific similarity metric efficiently, pioneering the LSH research for vector data of high dimensions (i.e., vector-to-vector search). In recent years, learning-based hashing methods for vector-to-vector search have achieved substantial progress [12]; in several tasks, these methods have been widely adopted such as large-scale visual search [13]–[15], object detection [16], [17], and machine learning [18]–[20].

The classic vector-to-vector nearest neighbor search has made substantial progress in hashing research. However, in LSH techniques research, few studies have been conducted for subspace-to-subspace search. If this problem is to be dealt with, it would be great to represent subspaces using compact signatures without losing the information of locality. Different from vector data, subspace data is more informative and complex because of a large amount of samples distributed in one subspace mainly. Meanwhile, despite that their dimensions may vary substantially, different subspaces also have inclusion relations. As a result, it is certainly difficult to distinguish similar subspaces and dissimilar subspaces, particularly among massive high dimensional databases with large cardinality.

[21] proposed the most famous pioneering method for the subspace search, which maps every subspace in $\mathbb{R}^n$ to the particular orthogonal projection matrix and further runs the nearest neighbor search equivalently over the projection matrix. Besides, the fast search with nice performance had been achieved in both [21] and [1]. Unfortunately, the computational cost of these methods is obviously expensive due to their heavy dependence on the product of the orthogonal projection matrix. A new LSH method was proposed in [2], by thresholding the angular mrtric for linear subspace. Unfortunately, as a result of the explosive growth of the number of possible proposals, the sublinear search is often subsequently downgraded to an exhaustive search at the querying stage.

Manuscript received December 9, 2019; revised February 11, 2020; accepted February 25, 2020. Date of publication March 2, 2020; date of current version December 17, 2020. This work was supported in part by the National Natural Science Foundation of China under Grants 61872021 and 61690202 and in part by Beijing Nova Program of Science and Technology under Grant Z191100001119050. The associate editor coordinating the review of this manuscript and approving it for publication was Dr. Jingdong Wang. *(Corresponding author: Xianglong Liu.)*

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Digital Object Identifier 10.1109/TMM.2020.2977459

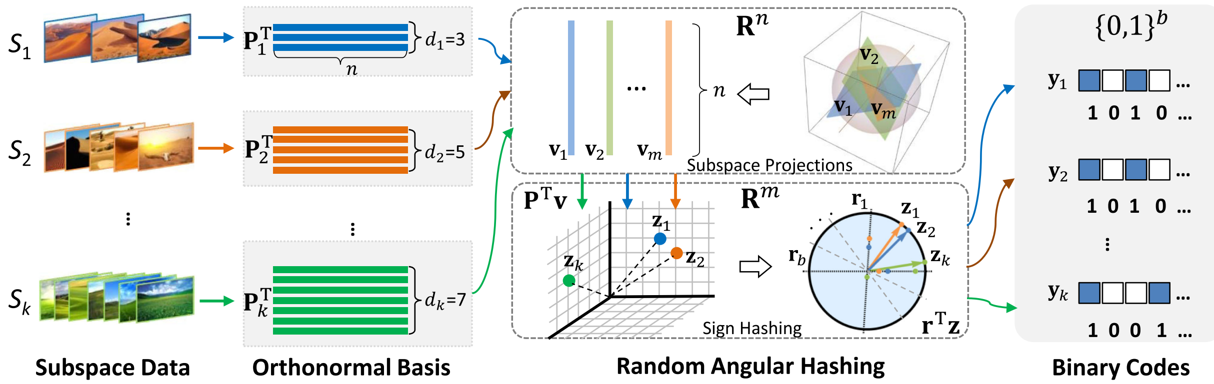


Fig. 1. Random angular hashing for subspace nearest neighbor search (*e.g.*, video sequences).

In light of the above, this paper proposes a novel and efficient locality-sensitive hashing over linear subspaces of random dimension for fast nearest neighbor search. The proposed approach uses a simple random projection paradigm, which is named random angular hashing (RAH), to get the subspace distribution characteristics efficiently by making use of the discriminative binary codes. In detail, the angle of subspaces, which is a basic subspace similarity metric, can be preserved mainly by RAH. We show how RAH produces binary codes for the video sequences in Fig. 1. As far as we know, the approach we proposed is the fastest subspace LSH proposed to date that still has powerful locality sensitivity for linear subspaces. Note that the whole paper extends upon a previous conference publication [22] which first proposed the idea of angular hashing. In this paper, we further present more detailed discussions about the related work and the theoretical analysis of the random angular hashing. Besides, we give the amplified experimental results over more tasks and more detailed discussions. On the whole, we theoretically and practically prove that the proposed method can simultaneously deliver discriminative and fast binary code generation in a way that outperforms existing state-of-the-art solutions.

We organize the paper as follows. Section II discusses the related work in vectorial hashing and subspace hashing. Section III and IV present the design details of the subspace angular hashing method and locality-sensitive angular hashing, respectively. The extensive experimental results and analyses are shown in Section V. Finally we conclude in Section VI.

II. Related Work

Subspaces often offer powerful representation in many tasks such as recognition, retrieval, and optimization. In these tasks, finding the nearest subspaces (i.e., subspace-to-subspace search) as a fundamental problem is often inevitably involved. Some examples of research work related to it include activity recognition [23], [24], video clustering [23], [25], pedestrian detection [26], face recognition [23], [27], [28], object recognition [28]–[30], feature representation [31], [32], gender recognition [33] and MRI data analysis [34]. [35] proposed a novel subspace clustering approach that segments the videos into consistent spatio-temporal regions with multiple classes, as a result, the common foreground has consistent labels across different videos. [36] introduced a subspace-based multimedia data mining framework for video semantic analysis, specifically video event/concept detection. For image classification, [37] studied discriminant subspace analysis as an adaptive method that can find an optimal discriminative subspace that can be adapted to different sample distributions. And in multi-class image classification, instead of learning a global subspace for all samples, [38] attempted to learn a local subspace for each sample mainly due to the label of each testing sample is only confused by a few classes which have very similar visual appearance to it.

Locality-Sensitive Hashing (LSH) [39] is one of the most famous hash-based nearest neighbor search approaches, which provides a very powerful technique for fast search over gigantic databases. In the literature for the vector and subspace data, there have been several well-studied methods. Among them, extensive efforts have been put on the vectorial hashing, while unfortunately, till now there have been very few studies focusing on the subspace hashing.

The essential difference between the vectorial hashing method and the subspace hashing method is the data they dealt with. The vectorial hashing method operates on the "vectorial data (i.e., vectors)" in the process of searching, while the subspace hashing method handles the subspace data. The different data types make their similarity metrics quite different. For example, the cosine distance and Euclidean distance are usually used for vector data, while the angular distance is commonly used for subspace data in the literature. Besides, in practice we usually concern much more about the hashing for subspace data of arbitrary dimension. This makes designing a locality sensitive subspace hashing much more difficult than vector, whose dimension is usually fixed.

A. Vectorial Hashing

For the high-dimensional vector data, to guarantee the search accuracy, the popular LSH method tries to embed similar data in original similarity metrics like l_p-normal ($p \in [0, 2]$) into similar codes by thresholding the random projections [40]. Unfortunately, since the projections are generated independently and randomly, usually long hash codes are required to meet the desired performance, which increases computation and memory consumption in practice. To generate compact, yet informative binary codes, various types of hashing algorithms have been further proposed following LSH [41]–[44], especially the learning based ones. For instance, [45] proposed a non-transitive hashing method which named Multi-Component Hashing, to identify the latent similarity components to cope with the non-transitive similarity relationships. [46] proposed a Relation-aware Heterogeneous Hashing, which provides a general framework for generating hash codes of data entities sitting in multiple heterogeneous domains. [47] proposed a hashing-based orthogonal deep model to learn accurate and compact multi-modal representations. [48] proposed a Locality Preserving Hashing method to improve the image retrieval performance. [49] introduced many of the efforts in sparsity-based heterogeneous feature selection, the related hashing index techniques, etc.

The vectorial hashing methods can also be adopted to address the nearest subspace search. This can be easily accomplished by first converting the subspace into the vector representation and then generating the hash codes using the vectorial hashing methods. [1] follows this paradigm, which adapts the subspace matrix into a high-dimensional vector, then applies the random hashing-based LSH to generate binary codes. There lies in the key point whether there exists a consistent similarity metrics between the subspace and the vector so that we can ensure the searched nearest vector corresponds to the nearest subspace. Fortunately, in [1] by transforming a linear subspace into a compact binary signature, and the Hamming distance between two signatures provides an unbiased estimate of the angular similarity between the two subspaces.

B. Subspace Hashing

Different from vectorial hashing, the subspace hashing directly maps the subspace data into the binary codes. There has been significant recent interest in linear subspace hashing problems in the past decade. The most well-known pioneering approach maps each subspace in $\mathbb{R}^n$ to its orthogonal projection matrix and subsequently performs the nearest neighbor search equivalently over the projection matrix [21]. This solution uniformly handles cases where the queries are points or subspaces, where query and database elements differ in dimensionality, and where the database contains subspaces of different dimensions. However, it has poor scaling in high dimensions, and may not guarantee sublinear query complexity. To address this issue, [2] provided a new LSH solution by simply thresholding the angular distance for linear subspace of arbitrary dimension, whose time complexity depends sub-linearly on the number of subspaces in the database. Moreover, considering that the subspace to point transforming process may cause subspace structural information loss, which influences the search accuracy, [50] proposed a new approach which learns the binary codes for subspaces following a similarity preserving criterion, and simultaneously leverages the learned binary codes to train matrix classifiers as hash functions. More recently, [51] proposed a novel ranking-based hashing framework that jointly learn two groups of linear subspaces, one for each modality, so that the features' ranking orders in different linear subspaces maximally preserve the cross-modal similarities.

III. Subspace Angular Hashing

Given a d-dimensional subspace $\mathrm{S} \subset \mathbb{R}^n$, we aim to seek out a locality-sensitive hash $\mathcal{H} : \mathbb{R}^{n\times d} \mapsto \{0,1\}^b$, i.e., one that can map any d-dimensional subspaces into a b-length binary code. In other words, two similar subspaces can be encoded by $\mathcal{H}$ with similar hash codes.

A. Angular Distance

Before we show the details of the subspace hash function we designed, the subspace distance metric commonly used in the literature will be first explained.

Given subspaces S_1 and S_2 with dimension d_1 and d_2 ($d_1 \leq d_2$) in $\mathbb{R}^n$, along with their orthonormal basis $\mathbf{P}_1$ and $\mathbf{P}_2$, their principal angles $\xi_1, \xi_2, ..., \xi_{d_1} \in [0, \frac{\pi}{2}]$ can be defined recursively as follows [52]:

$$\cos(\xi_i) = \max_{\mathbf{p}_i \in \mathrm{S}_1} \max_{\mathbf{q}_i \in \mathrm{S}_2} \left(\frac{\mathbf{p}_i^\top \mathbf{q}_i}{\|\mathbf{p}_i\|_2 \|\mathbf{q}_i\|_2} \right), \tag{1}$$

subject to

$$\mathbf{p}_i^\top \mathbf{p}_j = 0 \text{ and } \mathbf{q}_i^\top \mathbf{q}_j = 0, \text{ for } j = 1, 2, \ldots, i-1.$$

Note that each subspace S of d-dimension can be totally represented by its orthonormal basis $\mathbf{P} \in \mathbb{R}^{n\times d}$, which can be obtained efficiently by applying a Gram-Schmidt process with unit-normalization. Therefore, in the literature, the principal angle is naturally used to estimate the closeness of the subspaces, from the perspective of the angles between the orthonormal basis.

Based on the principal angles, the general angular distance can be defined as follows:

Definition 1 (Angular Distance): Given subspaces S_1 and S_2 in $\mathbb{R}^n$, with dimensions d_1 and d_2 ($d_1 \leq d_2$) respectively, the angle between S_1 and S_2 is $\xi_{\mathrm{S}_1,\mathrm{S}_2} = \arccos \frac{\sum_{i=1}^{d_1} \cos^2(\xi_i)}{\sqrt{d_1}\sqrt{d_2}}$, such that their angular distance can thus be defined as

$$\mathrm{dist}(\mathrm{S}_1, \mathrm{S}_2) = \frac{1}{\pi} \arccos \frac{\sum_{i=1}^{d_1} \cos^2(\xi_i)}{\sqrt{d_1}\sqrt{d_2}}. \tag{2}$$

Satisfying non-negativity, identity of indiscernible, inequality of triangle and symmetry, the angular distance is a "true" distance metric [1]. Further a metric space can be defined for all subspaces and a type of locality-sensitive hashing can be designed on the metric space later.

It should be noted here that it is often complex to directly compute the angular distance based on (1). However, we fortunately also know that [53]:

$$\|\mathbf{P}_1^\top \mathbf{P}_2\|_F^2 = \sum_{i=1}^{d_1} \cos^2(\xi_i). \tag{3}$$

Therefore, the computation of the angle $\xi_{\mathrm{S}_1,\mathrm{S}_2}$ can be simplified by $\|\mathbf{P}_1^\top \mathbf{P}_2\|_F^2$.

B. Random Subspace Projection

We point out that the orthonormal basis is not unique to one subspace. In order to distinguish a subspace quickly, it is highly critical that every subspace data should be represented uniquely and compactly.

Suppose both $\mathbf{P}$ and $\bar{\mathbf{P}}$ are the orthonormal basis for subspace S of d-dimension; then, given a unit vector $\mathbf{v} \in \mathbb{R}^n$, the following observation as lists

$$\|\mathbf{P}^\top \cdot \mathbf{v}\|_F^2 = \|\bar{\mathbf{P}}^\top \cdot \mathbf{v}\|_F^2. \tag{4}$$

This fact inspires us to design an α function that can represent the orthonormal matrix form of the subspace uniquely.

Definition 2 (α Function): A α function is defined based on a unit vector $\mathbf{v} \in \mathbb{R}^n$, mapping a d-dimensional subspace S in $\mathbb{R}^n$ to a real number in $[0, 1]$

$$\alpha_{\mathbf{v}}(\mathrm{S}) = \left\|\mathbf{P}_{\mathrm{S}}^\top \cdot \mathbf{v}\right\|_F^2 \tag{5}$$

where $\mathbf{P}_{\mathrm{S}}$ is an orthonormal matrix of S and $\|\cdot\|_F$ denotes the Frobenius norm.

Different orthonormal matrices have one α function for the same subspace, that is to say, the unique characteristics of the subspace data can be captured by the function.

Example 1: Let $\mathbf{P}_1 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$ and $\bar{\mathbf{P}}_1 = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ 0 & 0 \end{pmatrix}$ be the orthonormal matrice of subspace S_1 in $\mathbb{R}^3$, $\mathbf{P}_2 = \begin{pmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}$ be the orthonormal matrix of another subspace S_2 in $\mathbb{R}^3$. Then, for a unit vector $\mathbf{v} = (\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0)^\top$, we have

$$\|\mathbf{P}_1^\top \mathbf{P}_2\|_F^2 = \|\bar{\mathbf{P}}_1^\top \mathbf{P}_2\|_F^2 = 1, \text{ with } \xi_1 = 0, \xi_2 = \pi/2,$$
$$\alpha_{\mathbf{v}}(\mathrm{S}_1) = \|\mathbf{P}_1^\top \cdot \mathbf{v}\|_F^2 = \|\bar{\mathbf{P}}_1^\top \cdot \mathbf{v}\|_F^2 = 1,$$

and

$$\alpha_{\mathbf{v}}(\mathrm{S}_2) = \|\mathbf{P}_2^\top \cdot \mathbf{v}\|_F^2 = \frac{1}{2}.$$

Stimulated by the α function, the angular projection function can be defined for subspace S of d dimension, by attaching a proper basis $\alpha_0 d$, such that

$$z_{\mathbf{v}}(\mathrm{S}) = \alpha_{\mathbf{v}}(\mathrm{S}) + \alpha_0 d. \tag{6}$$

The projection function $z_{\mathbf{v}}$ is regarded as a helpful and smart representation of the subspace. Following we will indicate that as a result of that this kind of projection motivates a locality-sensitive hash function for the angular distance, the subspace will enjoy a similar hash code together with a similar ones. The following theorem illustrates that $z_{\mathbf{v}}$ powerfully connects the angular distance.

Theorem 1: Given subspaces S_1 and S_2 in $\mathbb{R}^n$, with dimension d_1 and d_2 respectively, and

$$\alpha_0 = \frac{\sqrt{2}}{\sqrt{n^3 + 2n^2}} - \frac{1}{n}, \tag{7}$$

if $\mathbf{v}$ is a stochastic variable vector evenly distributed in the unit sphere of $\mathbb{R}^n$, then the equations are as follows

$$\mathbf{E}\left[z_{\mathbf{v}}(\mathrm{S}_1) z_{\mathbf{v}}(\mathrm{S}_2)\right] = \frac{2}{(n+2)n} \left\|\mathbf{P}_1^\top \mathbf{P}_2\right\|_F^2, \tag{8}$$

where $\mathbf{P}_1$ and $\mathbf{P}_2$ are any orthonormal basis of subspaces S_1 and S_2.

Proof: Writing $\mathbf{P}_1, \mathbf{P}_2$ in the following forms:

$$\mathbf{P}_1 = (\mathbf{a}_1, \mathbf{a}_2, \ldots \mathbf{a}_{d_1}), \mathbf{P}_2 = (\mathbf{b}_1, \mathbf{b}_2, \ldots \mathbf{b}_{d_2}) \tag{9}$$

and denoting the one-dimensional subspace spanned by $\mathbf{a}_i, \mathbf{b}j$ as $\mathrm{A}_i, \mathrm{B}_j$, it is clear to see that

$$\begin{aligned} z_{\mathbf{v}}(\mathrm{S}_1) &= \|\mathbf{P}_1^\top \mathbf{v}\|_F^2 + d_1\alpha_0 \\ &= \Sigma_{i=1}^{d_1}(\|\mathbf{a}_i^\top \mathbf{v}\|_F^2 + \alpha_0) \\ &= \Sigma_{i=1}^{d_1} z_{\mathbf{v}}(\mathrm{A}_i) \end{aligned} \tag{10}$$

$$\begin{aligned} z_{\mathbf{v}}(\mathrm{S}_2) &= \|\mathbf{P}_2^\top \mathbf{v}\|_F^2 + d_2\alpha_0 \\ &= \Sigma_{j=1}^{d_2}(\|\mathbf{b}_j^\top \mathbf{v}\|_F^2 + \alpha_0) \\ &= \Sigma_{j=1}^{d_2} z_{\mathbf{v}}(\mathrm{B}_j) \end{aligned} \tag{11}$$

So,

$$\begin{aligned} &\mathbf{E}\left[(z_{\mathbf{v}}(\mathrm{S}_1) z_{\mathbf{v}}(\mathrm{S}_2)\right] \\ &= \mathbf{E}(\Sigma_{i=1}^{d_1} z_{\mathbf{v}}(\mathrm{A}_i) \Sigma_{j=1}^{d_2} z_{\mathbf{v}}(\mathrm{B}_j)) \\ &= \mathbf{E}(\Sigma_{i=1}^{d_1} \Sigma_{j=1}^{d_2} (z_{\mathbf{v}}(\mathrm{A}_i) z_{\mathbf{v}}(\mathrm{B}_j))) \\ &= \Sigma_{i=1}^{d_1} \Sigma_{j=1}^{d_2} \mathbf{E}(z_{\mathbf{v}}(\mathrm{A}_i) z_{\mathbf{v}}(\mathrm{B}_j)) \end{aligned} \tag{12}$$

According to (3),

$$\begin{aligned} &\Sigma_{i=1}^{d_1} \Sigma_{j=1}^{d_2} \mathbf{E}(z_{\mathbf{v}}(\mathrm{A}_i) z_{\mathbf{v}}(\mathrm{B}_j)) \\ &= \Sigma_{i=1}^{d_1} \Sigma_{j=1}^{d_2} \frac{2}{(n+2)n} \|\mathbf{a}_i^\top \mathbf{b}_j\|_F^2 \\ &= \frac{2}{(n+2)n} \Sigma_{i=1}^{d_1} \Sigma_{j=1}^{d_2} \|\mathbf{a}_i^\top \mathbf{b}_j\|_F^2 \\ &= \frac{2}{(n+2)n} \|\mathbf{P}_1^\top \mathbf{P}_2\|_F^2 \end{aligned} \tag{13}$$

Note that the projection value of $\alpha_{\mathbf{v}}$ function highly depends on the dimension of the input subspace. Basically, the higher dimension will get higher projection value. To support the comparison between subspaces of arbitrary dimension, we need to introduce a bias, which can help eliminate the effect of dimension size. Therefore, we introduce the proper bias $\alpha_0 d$ in $z_{\mathbf{v}}$. Since α is negative for $n > 0$, the bias is negatively correlated with the dimension d of the subspace. Therefore, we can easily conclude that the bias actually makes the $z_{\mathbf{v}}$ projection of subspace with the higher dimension comparable with that with the lower dimension.

As Equation (3) declears, the angular distance between two subspaces has a inseparable relationship with the Frobenius norm of their orthonormal matrices product. The theoretical outcome here further demonstrates that the product of their projection values can be regarded as equivalent to the Frobenius norm. In short, we can conclude that

$$\mathbf{E}\left[z_{\mathbf{v}}(\mathrm{S}_1) z_{\mathbf{v}}(\mathrm{S}_2)\right] = \frac{2}{(n+2)n} \sum_{i=1}^{d_1} \cos^2(\xi_i). \tag{14}$$

We find out that the projection function $z_{\mathbf{v}}$ has the unexpected performance of be capable of maintain the angular distances. ■

Example 2: Let $\mathbf{P}_1 = (1, 0)^\top$ and $\mathbf{P}_2 = (\cos\gamma, \sin\gamma)^\top$ (γ is constant) respectively be the orthonormal matrix of subspace S_1 and S_2 in $\mathbb{R}^2$, with dimension $d_1 = d_2 = 1$. Then, for a unit vector $\mathbf{v} = (\cos\tau, \sin\tau)^\top$, $\tau \sim \mathrm{U}[0, 2\pi]$, we have

$$\mathbf{E}[\alpha_{\mathbf{v}}(\mathrm{S}_1)\alpha_{\mathbf{v}}(\mathrm{S}_2)] = \frac{\cos^2(\gamma)}{4} + \frac{1}{8}, \tag{15}$$

and

$$\mathbf{E}[\alpha_{\mathbf{v}}(\mathrm{S}_2)] = \frac{1}{2},\ \mathbf{E}[\alpha_{\mathbf{v}}(\mathrm{S}_1)] = \frac{1}{2}.$$

Thus, with $\alpha_0 = \frac{\sqrt{2}}{4} - \frac{1}{2}$ and $\xi_1 = \gamma$,

$$\begin{aligned}\mathbf{E}[z_{\mathbf{v}}(\mathrm{S}_1)z_{\mathbf{v}}(\mathrm{S}_2)] &= \mathbf{E}[\alpha_{\mathbf{v}}(\mathrm{S}_1)\alpha_{\mathbf{v}}(\mathrm{S}_2)] \\ &\quad + \alpha_0\left(\mathbf{E}[\alpha_{\mathbf{v}}(\mathrm{S}_1)] + \mathbf{E}[\alpha_{\mathbf{v}}(\mathrm{S}_2)]\right) + \alpha_0^2 \\ &= \frac{\cos^2(\xi_1)}{4}.\end{aligned}$$

To completely represent the subspace largely, a series of random projection function $\{z_{\mathbf{v}_1}, z_{\mathbf{v}_2}, \ldots, z_{\mathbf{v}_m}\}$ can be utilized as one composite projection functions $\mathcal{Z}$ and the expectation can be approximated in (14) by taking the average. More especially, every subspace S can be represented as a m-dimensional vector $\mathbf{z}_{\mathrm{S}}$ using projection $\mathcal{Z}$:

$$\mathbf{z}_{\mathrm{S}} = \mathcal{Z}(\mathrm{S}) \doteq \left(z_{\mathbf{v}_1}(\mathrm{S}), z_{\mathbf{v}_2}(\mathrm{S}), \ldots, z_{\mathbf{v}_m}(\mathrm{S})\right)^\top. \tag{16}$$

It is also therefore simply to find out that $\lim_{m\to+\infty} \frac{1}{m}\mathbf{z}_{\mathrm{S}_1}^\top \mathbf{z}_{\mathrm{S}_2} = \mathbf{E}[z_{\mathbf{v}}(\mathrm{S}_1)z_{\mathbf{v}}(\mathrm{S}_2)]$.

Using the central limit theorem, such that when m is large enough, $\frac{1}{m}\mathbf{z}_{\mathrm{S}_1}^\top \mathbf{z}_{\mathrm{S}_2}$ approximately obeys the normal distribution $\mathcal{N}(\mu, \frac{\sigma^2}{m})$

$$\frac{1}{m}\mathbf{z}_{\mathrm{S}_1}^\top \mathbf{z}_{\mathrm{S}_2} \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{m}\right) \tag{17}$$

with $\mu = \mathbf{E}[z_{\mathbf{v}}(\mathrm{S}_1)z_{\mathbf{v}}(\mathrm{S}_2)]$ and $\sigma = \sqrt{\mathbf{V}[z_{\mathbf{v}}(\mathrm{S}_1)z_{\mathbf{v}}(\mathrm{S}_2)]}$, where $\mathbf{V}[z_{\mathbf{v}}(\mathrm{S}_1)z_{\mathbf{v}}(\mathrm{S}_2)]$ is the variance.

Note that $\mathbf{V}[z_{\mathbf{v}}(\mathrm{S}_1)z_{\mathbf{v}}(\mathrm{S}_2)]$ is bounded for any $\mathrm{S}_1, \mathrm{S}_2$. Therefore, we obtain that

$$\lim_{m\to+\infty} \frac{\sigma}{\sqrt{m}} = 0, \tag{18}$$

which means that the approximation loss can be controlled by the length m of the angular projection. When m is sufficiently large, we can guarantee that the approximation nearly converges.

The projection vector owns powerful ability of both approximating and fast computation. It can be guaranteed that the approximation nearly converges when m is large enough.

IV. Locality-Sensitive Subspace Hashing

The random subspace projection can efficiently produce the representation of the vector. Unfortunately, in order to ensure a excellent approximation of the subspace similarities, it is usually necessary to rely on lots of projections, which generate a big vectorial representation. Unfortunately, The method results in high computational costs in searching the nearest subspace. We aim to construct a new family of locality-sensitive hash, which can further encode the vector representation of each subspace to a compact binary code and thus help us complete fast search without too much memory consumption.

A. Sign Random Hashing

To achieve this goal, we further adopt the popular sign random hashing technique [11], which historically has been widely and successfully used to preserve the angular distance among the high-dimensional data, using the binary codes with a large probability. This simple process, fortunately, forms a new kind of locality-sensitive hash (LSH) for subspace data, which can generate the binary codes by encoding the vectorial representation to accelerate the search faithfully, meanwhile the locality is preserved among the subspaces.

Definition 3 (Sign Random Hashing): A sign random hashing $f_{\mathbf{r}}(\cdot)$, mapping a vector $\mathbf{x} \in \mathbb{R}^m$ to a binary bit using $\mathbf{r} \in \mathbb{R}^m$, is defined as

$$f_{\mathbf{r}}(\mathbf{x}) = \mathrm{sgn}(\mathbf{r}^\top \mathbf{x}). \tag{19}$$

For any two vectors $\mathbf{x}$ and $\mathbf{y}$, if $\mathbf{r}$ is a random variable vector, uniformly distributed in the unit sphere of $\mathbb{R}^m$, we can conduct that [54]

$$\Pr[f_{\mathbf{r}}(\mathbf{x}) = f_{\mathbf{r}}(\mathbf{y})] = 1 - \frac{1}{\pi}\arccos\left(\frac{\mathbf{x}^\top \mathbf{y}}{\|\mathbf{x}\|_2\|\mathbf{y}\|_2}\right). \tag{20}$$

In order to solve the subspace search problem, a metric space (Ω, dist) can further be defined for all subspaces of $\mathbb{R}^n$ based on the angular distance for subspaces in Equation (2), where $\Omega = \{\mathrm{S} \subset \mathbb{R}^n | \mathrm{S}\text{ is a subspace of }\mathbb{R}^n\}$. We then discover that the sign random hashing which is combined with the random subspace projection function can be regarded as a new locality-sensitive hash function in the metric space (Ω, dist). And the following theorem holds:

Theorem 2: When m is sufficiently large, $\mathcal{H} = f_{\mathbf{r}} \circ \mathcal{Z}$ is a locality-sensitive hash defined on (Ω, dist) approximately, i.e., $\mathcal{H}$ is $(\delta, \delta(1+\epsilon), 1-\delta, 1-\delta(1+\epsilon))$-sensitive with $\delta, \epsilon > 0$.

Proof: According to Theorem 1, for any two subspaces S_1 and S_2 in $\mathbb{R}^n$, with dimensions d_1 and d_2 ($d_1 \le d_2$) respectively, we have

$$\lim_{m\to+\infty} \frac{1}{m}\mathbf{z}_{\mathrm{S}_1}^\top \mathbf{z}_{\mathrm{S}_2} = \frac{2}{n(n+2)}\sum_{i=1}^{d_1} \cos^2(\xi_i),$$

$$\lim_{m\to+\infty} \frac{1}{m}\|\mathbf{z}_{\mathrm{S}_1}\|_2^2 = \frac{2}{n(n+2)} d_1,$$

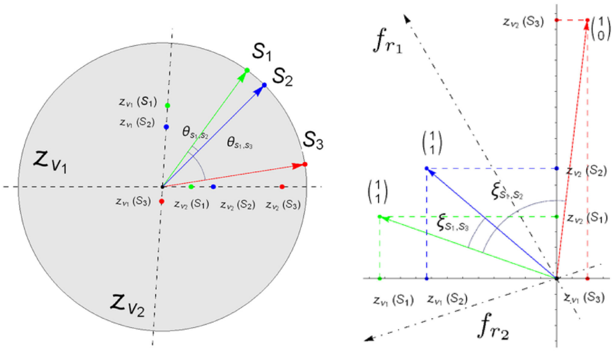
$$\lim_{m\to+\infty} \frac{1}{m}\|\mathbf{z}_{\mathrm{S}_2}\|_2^2 = \frac{2}{n(n+2)} d_2.$$

Therefore,

$$\lim_{m\to+\infty} \frac{\mathbf{z}_{\mathrm{S}_1}^\top \mathbf{z}_{\mathrm{S}_2}}{\|\mathbf{z}_{\mathrm{S}_1}\|_2\|\mathbf{z}_{\mathrm{S}_2}\|_2} = \frac{\sum_{i=1}^{d_1}\cos^2(\xi_i)}{\sqrt{d_1}\sqrt{d_2}},$$

when m is large enough, we approximately have

$$\frac{\mathbf{z}_{\mathrm{S}_1}^\top \mathbf{z}_{\mathrm{S}_2}}{\|\mathbf{z}_{\mathrm{S}_1}\|_2\|\mathbf{z}_{\mathrm{S}_2}\|_2} \approx \frac{\sum_{i=1}^{d_1}\cos^2(\xi_i)}{\sqrt{d_1}\sqrt{d_2}}$$



(a) Random Subspace Projection (b) Binary Codes Generation

Fig. 2. Procedure of random angular hashing.

TABLE I
COMPARISON OF COLLISION PROBABILITY p AND THE COMPLEXITY t OF ENCODING A SUBSPACE FROM ORTHONORMAL BASIS TO A b-LENGTH BINARY CODE

	RAH	BSS	GLH
p	$1-\delta$	$1-\delta$	-
t	$O(bm+dmn)$	$O(bn^2)$	$O(bdn)$

Algorithm 1: Random Angular Hashing

Input: A d-dimensional subspace $\mathrm{S} \subset \mathbb{R}^n$;
Get the orthonormal basis $\mathbf{P}_s^T \in \mathbb{R}^{n*d}$ for S;
Randomly choose a serials of unit vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_m \in \mathbb{R}^n$;
for $j = 1, \ldots, m$ **do**
  Calculate $\alpha_{\mathbf{v_j}}(\mathrm{S}) = \|\mathbf{P}_{\mathrm{S}}^\top \cdot \mathbf{v_j}\|_F^2$;
  Calculate $z_{\mathbf{v_j}}(\mathrm{S}) = \alpha_{\mathbf{v}}(\mathrm{S}) + \alpha_0 d$, $\mathbf{v} = \mathbf{v}_j$, using α_0 in Eq. (7).
end for
Generate an m-dimensional vectorial representation $\mathbf{z}_{\mathrm{S}}$, using Eq. (16).
Randomly choose a serials of unit vectors $\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_b$, ($\mathbf{r}$ has the same dimensions with the vector $\mathbf{z}_{\mathrm{S}}$);
for $t = 1, \ldots, b$ **do**
  $\mathbf{y}_t = \mathrm{sgn}\,(\mathbf{r}_t^T \cdot \mathbf{z}_{\mathrm{S}})$, $\mathbf{y}_t$ means the t-th bit of the binary code of $\mathbf{y}$;
end for
Output: the binary codes $\mathbf{y} = [\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_b]$.

and

$$\arccos \frac{\mathbf{z}_{\mathrm{S}_1}^\top \mathbf{z}_{\mathrm{S}_2}}{\|\mathbf{z}_{\mathrm{S}_1}\|_2 \|\mathbf{z}_{\mathrm{S}_2}\|_2} \approx \arccos \frac{\sum_{i=1}^{d_1} \cos^2(\xi_i)}{\sqrt{d_1}\sqrt{d_2}}.$$

Therefore, the following holds:

$$\begin{aligned}
&\Pr[f_{\mathbf{r}} \circ \mathcal{Z}(\mathrm{S}_1) = f_{\mathbf{r}} \circ \mathcal{Z}(\mathrm{S}_2)] \\
&= 1 - \frac{1}{\pi} \arccos \left(\frac{\mathbf{z}_{\mathrm{S}_1} \cdot \mathbf{z}_{\mathrm{S}_2}}{\|\mathbf{z}_{\mathrm{S}_1}\|_2 \|\mathbf{z}_{\mathrm{S}_2}\|_2} \right) \\
&\approx 1 - \frac{1}{\pi} \arccos \frac{\sum_{i=1}^{d_1} \cos^2(\xi_i)}{\sqrt{d_1}\sqrt{d_2}} \\
&= 1 - \mathrm{dist}(\mathrm{S}_1, \mathrm{S}_2)
\end{aligned}$$

If $\mathrm{dist}(\mathrm{S}_1, \mathrm{S}_2) \le \delta$, we then have

$$\Pr[f_{\mathbf{r}} \circ \mathcal{Z}(\mathrm{S}_1) = f_{\mathbf{r}} \circ \mathcal{Z}(\mathrm{S}_2)] \ge 1 - \delta = p_1$$

Likewise, when $\mathrm{dist}(\mathrm{S}_1, \mathrm{S}_2) > \delta(1+\epsilon)$, we have

$$\Pr[f_{\mathbf{r}} \circ \mathcal{Z}(\mathrm{S}_1) = f_{\mathbf{r}} \circ \mathcal{Z}(\mathrm{S}_2)] < 1 - \delta(1+\epsilon) = p_2,$$

where $p_2 < p_1$. This completes the proof. ■

We now have our random angular hashing (RAH) as a new type of locality sensitive hash for subspace, and the theoretical outcome states clearly that the combined function $\mathcal{H}$ is able to satisfy the locality sensitivity for the angular distance, when making use of massive angular projections. The following two steps are used by the LSH function for the subspace: the first step is to project the subspace into a high-dimensional vector and the second step is to encode the vector into a binary code, as Fig. 2 shows. Table I shows the result of comparison of the locality sensitivity and time complexity state-of-the-art methods. Compared to $O(n^2)$ of existing subspace hashing approaches [1], the time complexity of the RAH has a linear relationship to the space dimension n, which decreases the computation required largely.

With the LSH property, following [55], [56] the sublinear search time based on the binary codes generated for all subspaces can be theoretically guaranteed.

Theorem 3: Given a subspace database $\mathcal{D}$ with N subspaces in $\mathbb{R}^n$, for a subspace query S, if there exists a subspace S^* such that $\mathrm{dist}(\mathrm{S}, \mathrm{S}^*) \le \delta$, then with $\rho = \frac{\ln p_1}{\ln p_2}$ (1) using N^ρ hash tables with $\log_{1/p_2} N$ hash bits, the random angular hash of an even order is able to return a database subspace $\bar{\mathrm{S}}$ such that $\mathrm{dist}(\mathrm{S}, \bar{\mathrm{S}}) \le \delta(1+\epsilon)$ with probability of at least $1 - \frac{1}{c} - \frac{1}{e}, c \ge 2$; (2) the query time is sublinear to the entire data number N, with $N^\rho \log_{1/p_2} N$ bit generations and cN^ρ pairwise distances computation.

On the whole, we illustrate the procedure of the locality sensitive subspace hashing for subspaces of arbitrary dimension in Fig. 1 and Algorithm 1. The first step is to get the orthonormal basis for a subspace data, and as a result, a subspace of d-dimension can be totally represented by its orthonormal basis $\mathbf{P} \in \mathbb{R}^{n*d}$. Then the second step is to choose a serial of unit vectors randomly to map the subspace into a serial of corresponding numbers in [0, 1]. And finally, the angular projection of the subspace $\mathbf{z}_{\mathrm{v}}$ is to be got stimulated by the mapping function, outputting the binary codes through the operation of the sign function.

B. Balanced Binary Coding

In order to complete the fast search over the massive subspace database, using our RAH based hash function $\mathcal{H}$ we can turn the subspace data into binary codes and quickly approximate their similarities based on Hamming distances. In practice, the balanced binary codes can help reach satisfying performance, owning to the uniform distribution of the subspace data in the Hamming space.

To achieve this goal, in practice we can simply adopt the balanced processing that subtracts the mean value from the angular hashing vector for each subspace. This can be easily done by appending a bias to the sign random function as follows

$$f_{\mathbf{r}}^{+}(\mathbf{z}_{\mathrm{S}}) = \mathrm{sgn}(\mathbf{r}^{\top}\mathbf{z}_{\mathrm{S}} - c_0) \tag{21}$$

where c_0 serves as the mean value of the hashing over all $\mathbf{z}_{\mathrm{S}}$. Practically, we can estimate the mean from a small subspace set $\{\mathrm{S}_i\}(1 \leq i \leq N_s)$ with N_s ($N_s \ll N$) subspaces. In this case, we have an improved hash function $\mathcal{H}^{+} = f_{\mathbf{r}}^{+} \circ \mathcal{Z}$. With the balanced processing, we can keep the balance of $\{0, 1\}$ generated by $f_{\mathbf{r}}^{+}(\cdot)$. Such kind of processing has been widely applied in many applications. It should be emphasized that the balanced processing might break the locality sensitivity, since it is not an angle-invariant transformation. However, we practically find that it owns strong capability to preserve the angular distances.

V. Experiments

The subspace search has been adopted in a wide range of applications, including face recognition, action recognition, video search, etc. In order to make a comprehensive evaluation on our RAH method, three of the most popular tasks are chosen: namely, face recognition, video retrieval, and gesture recognition. The essences of the three tasks, i.e., face recognition, video retrieval, and gesture recognition, are the same, namely, in these tasks subspace search had been widely and successfully adopted. The key difference in them is that they treat different entities as the subspaces. In the face recognition task, face images with certain pose under different illumination conditions lie the near linear subspaces of "low dimension". As a result, a subspace can represent a set of face images belonging to the same person easily, so that we can regard the face recognition in this scenario as an example of the nearest subspace search. In the video retrieval task, the consecutive frames can be regarded as a single subspace, which naturally encodes the temporal relations among frames. The gesture recognition task is much like the video retrieval task and the face recognition task because it involves discrimination based on several similar successive images and each set of consecutive images can be represented by a subspace, while the recognition can be completed based on the nearest subspace search.

Our RAH method has been compared with a number of state-of-the-art subspace hashing approaches, including BHZM [21], Binary Signatures for Subspaces (BSS) [1] and Grassmannian-based Locality Hashing (GLH) [2]. To generate hash codes, all these methods base on the random way. Also, several state-of-the-art unsupervised vector hashing approaches are adopted as the baselines, including random locality sensitive hashing (LSH) [11], and several approaches base on learning: namely, dubbed discrete proximal linearized minimization (DPLM) [57], iterative quantization (ITQ) [14], and adaptive binary quantization (ABQ) [58].

The accuracy of searching and computational efficiency is undoubtedly important for hashing methods, so we evaluate the hashing performance in terms of the precision and the search time for the three subspace search tasks. Specifically, for all tasks, we report the search time per query, which is widely adopted in hashing research. For face recognition and gesture recognition, since usually users mainly concern the top-ranked searched results, especially the top-1 results, we report the top-1 precision. For the video searching task, which is a kind of information retrieval task, where the common evaluation metric is the mean average precision (MAP) that takes both precision and recall into consideration.

A. Face Recognition

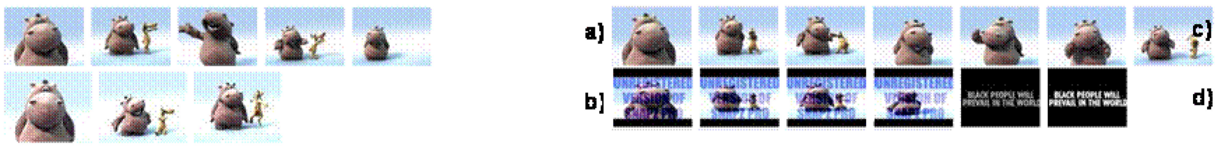
Face images with certain pose under different illumination conditions lie near linear subspaces of low dimension [1]. Accordingly, as a result, a subspace can represent a set of face images belonging to the same person easily, we can regard the face recognition in this scenario as an example of nearest subspace search.

Datasets: Two popular datasets are utilized here: the Extended Yale Face Database B [59], containing 38 individuals, each with about 64 near-frontal images under different illuminations [Fig. 3(a)], and the PIE database [60], consisting of 68 individuals, each with 170 images under five near-frontal poses, 43 different illumination conditions and four different expressions. On the PIE database, the frontal face images (C26 and C27) are chosen in this work. On the Extended Yale Face Database B, 30 images per individual as the database set are chosen randomly, while 11 images are chosen from the remainder as the testing set. Similarly, the database and testing set contains 20 and 15 images per individual respectively on the PIE database.

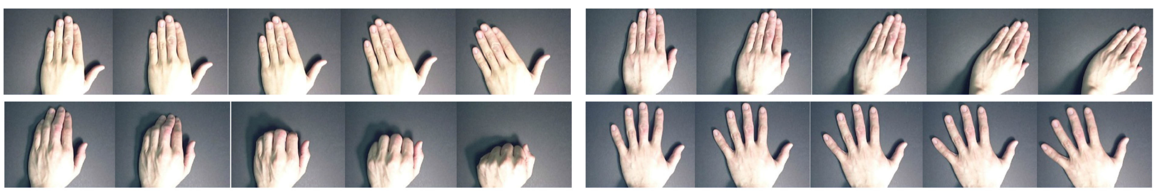
Experiment Setup: Each face image generates a 1,024-length vectorial representation; thus, a subspace in $\mathbb{R}^{1024}$ will be constituted by images of the same individual. For each individual, a representative subspace of dimension $d = 9$ on Yale Face and $d = 14$ for PIE datasets is extracted by taking the top main parts, as BSS did. This is mainly because compared with the Yale Face Database B, the PIE database, consisting of 68 individuals, each with 170 images under five nearfrontal poses, 43 different illumination conditions, and 4 different expressions, are more complex. For the testing set, moreover, the query subspaces are generated with different dimensions ($d_q = 7, 9, 11$ on Yale Face and $d_q = 13, 14, 15$ on PIE respectively), aiming to cover the typical cases: i.e., the dimension of query subspace is higher, lower than and equal to that of the database, so that we can comprehensively evaluate the encoding power of different hashing methods with respect to different dimensions. For the vector hashing approaches, we use the original face vectors directly without reduction. At the testing stage, for each query we search from the database for the subspaces with the nearest binary codes to the query based on the Hamming distance.

Results: The precision of the top-1 results and the search time of the different approaches for the face recognition task are listed in Tables II and III. Here, the search time consists of two sections: namely, the encoding time and the retrieval time

(a) The Extended Yale Face Database B



(b) UQ_VIDEO database



(c) Cambridge Gestures database

Fig. 3. The examples of face recognition, video retrieval and gesture datasets.

TABLE II
FACE RECOGNITION PERFORMANCE ON YALE FACES

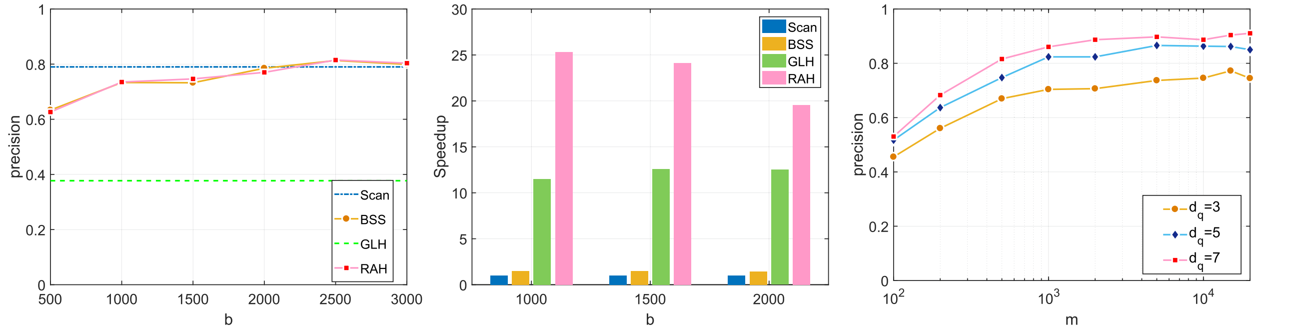
	METHODS	PRECISION (%), $b = 512, d = 9, m = 10^4$			SEARCH TIME (S)		
		$d_q = 7$	$d_q = 9$	$d_q = 11$	$d_q = 7$	$d_q = 9$	$d_q = 11$
YALE FACE	LSH		45.84 $\pm$3.49			0.0025	
	ITQ		47.89 $\pm$3.86			0.0016 (TRAIN TIME 5.73)	
	DPLM		46.84 $\pm$5.10			0.0022 (TRAIN TIME 0.49)	
	ABQ		58.42 $\pm$1.05			0.0028 (TRAIN TIME 22.26)	
	BHZM	14.21 $\pm$5.41	16.31 $\pm$8.55	16.84 $\pm$3.93	0.0080	0.0080	0.0087
	BSS	74.21 $\pm$8.05	72.63 $\pm$4.27	73.68 $\pm$3.72	0.1349	0.1349	0.1219
	RAH	72.10 $\pm$6.13	**73.10** $\pm$8.58	**82.10** $\pm$6.09	0.0078	0.0078	0.0084

TABLE III
FACE RECOGNITION PERFORMANCE ON PIE FACES

	METHODS	PRECISION (%), $b = 1024, d = 14, m = 10^4$			SEARCH TIME (S)		
		$d_q = 13$	$d_q = 14$	$d_q = 15$	$d_q = 13$	$d_q = 14$	$d_q = 15$
PIE FACE	LSH		57.64 $\pm$2.35			0.0075	
	ITQ		52.64 $\pm$1.44			0.0077 (TRAIN TIME 13.48)	
	DPLM		79.41 $\pm$2.07			0.0079 (TRAIN TIME 6.82)	
	ABQ		81.47 $\pm$1.99			0.0137 (TRAIN TIME 3.17)	
	BHZM	38.82 $\pm$4.97	41.76 $\pm$2.72	37.94 $\pm$5.76	0.0166	0.0175	0.0186
	BSS	97.05 $\pm$2.63	95.58 $\pm$2.27	95.29 $\pm$2.35	0.1252	0.1274	0.1259
	RAH	96.17 $\pm$0.72	**96.47** $\pm$0.72	**97.35** $\pm$2.35	0.0089	0.0111	0.0114

per query subspace. In the table, the learning-based vector hashing approaches have a better performance, compared with the random LSH approaches, however, the learning-based vector hashing approaches also rely heavily on the training process, which requires extra computational cost (see the training time). Moreover, we can also draw the conclusion that the random subspace hashing approaches RAH and BSS outperform much than all vector hashing approaches: this is because the vector hashing, which takes the vector data into account independently, lacking the ability to capture the intrinsic distributions in the subspace. In all cases, our RAH performance is close to and usually even better than that of BSS; this is consistent with our analysis in Table I that RAH and BSS have the same locality sensitivity ability to the angular distance. As for the query time, RAH can obtain up to 16× speedup, compared to BSS.

Fig. 4(a) and (b) show the study of the effect of the code length b on the precision performance and the speed acceleration (i.e., speedup). Here, we randomly choose d and d_q of each subspace from $\{3, 5, 7\}$, trying to simulate a real-world case. The performance of GLH is not affected by the code length, as a result of that GLH works in a different way from BSS and RAH in this case. In the figures, we also present the results of using the



(a) Precision w.r.t. b (b) Speedup w.r.t. b (c) RAH Precision w.r.t. m

Fig. 4. The effect of different b and m on Yale Face. To simulate real-world scenarios, we adopt random d and d_q in (a) and (b), and random d in (c).

TABLE IV
VIDEO RETRIEVAL PERFORMANCE ON UQ_VIDEO WITH EXPERIMENTAL SETTING I

METHODS	MAP (%), $d = d_q = 5$			SEARCH TIME (S)		
	$b = 64$	$b = 128$	$b = 256$	$b = 64$	$b = 128$	$b = 256$
LSH	53.89 ± 2.75	55.71 ± 2.93	55.76 ± 0.29	0.3846	0.5213	0.7405
ITQ	49.58 ± 1.16	50.68 ± 0.94	-	0.3865	0.5178	-
DPLM	54.28 ± 1.28	61.76 ± 1.46	64.38 ± 1.03	0.3852	0.5268	0.7158
ABQ	36.18 ± 3.57	36.26 ± 1.35	38.74 ± 0.29	0.5217	0.7404	1.3365
BSS	53.28 ± 1.88	65.31 ± 0.70	76.31 ± 1.12	0.0685	0.0685	0.0779
RAH	**54.92** ± 2.99	**68.09** ± 2.15	**77.50** ± 1.18	0.0721	0.0753	0.0892

exhaustive search ('Scan'). From these figures, we can clearly draw the conclusion that all the subspace hashing approaches develop the search efficiency clearly, while our RAH achieves similar performance to BSS, due to the same collision probability. Moreover, RAH is able to approximate or even outperform the exhaustive search strategy, while also completing the search at a much higher speed (more than a $10\times$ speedup relative to BSS and Scan), with more hash bits.

Fig. 4(c) illustrates the performance of RAH about the dimension m of the subspace projection vector (here we set $b = 2,000$). A larger m guarantees a better performance generally, which is consistent with our approximation analysis in Section III-B. We can see from these results that $m > 500$ is large enough for satisfactory performance practically.

B. Near-Duplicated Large-Scale Video Retrieval

The evaluation of the performance in the video retrieval task is introduced next. Here, thinking poorly of the temporal relations among frames, traditional vector hashing methods encode each video frame into a binary code individually. Fortunately, the subspace hashing can solve this problem by treating the consecutive frames as a single subspace.

Datasets: The UQ_VIDEO [15] is one of the largest databases for near-duplicate video retrieval, containing 169,952 videos totally and 3,305,525 key frames extracted from these videos [Fig. 3(b)]. 24 picked query videos are offered in the database, each of which has several near-duplicate videos, while other videos offer as background videos. The offered global color histogram feature with 162 dimensions is used as the input feature for each frame.

Experiment Setup I: In this experiment, we first design the database set with more than 10,000 videos, rather than using the whole database, and further remove those videos with very few frames. The database primarily consists of two parts: all the videos (more than 3,000) belonging to the 24 classes consistent with the given 24 queries, and the randomly selected background videos. For each video, every $d = 5$ consecutive frames without overlap are used to form a subspace. At the search stage, according to the averaged Hamming distances to the query, we use the Hamming distance ranking strategy to rank all video clips in the database set.

Results: Table IV shows the mean average precision (MAP) and search time using Hamming distance ranking in the video search task. The similar results are obtained to those which obtained on the whole dataset: in short, the subspace hashing methods still perform far better than the state-of-the-art vector hashing methods. Moreover, when a different number of hash bits are used, our RAH method can have the best performance consistently and is more than $8\times$ faster than the random vector hashing method LSH.

Experiment Setup II: We further form a database set with nearly 3,000,000 key frames to construct the database set containing all videos except those with very few frames. We use every $d = 5$ consecutive frames without overlap to form a subspace for each video. We use the Hamming distance ranking strategy to rank all the video clips in the database set according to the averaged Hamming distances to the query at the search stage.

Results: Table V shows the mean average precision (MAP) and search time using Hamming distance ranking in the video search task. We can easily see from these results that the subspace hashing methods achieve far better performance than the

TABLE V
VIDEO RETRIEVAL PERFORMANCE ON UQ_VIDEO WITH EXPERIMENTAL SETTING II

METHODS	MAP (%), $d = d_q = 5$			SEARCH TIME (S)		
	$b = 128$	$b = 256$	$b = 512$	$b = 128$	$b = 256$	$b = 512$
LSH	30.73 ± 1.01	30.20 ± 0.34	31.53 ± 0.06	5.5660	7.5578	16.7660
ITQ	28.92 ± 0.30	-	-	5.3990	-	-
DPLM	34.09 ± 1.60	37.24 ± 0.95	39.15 ± 0.91	5.0560	6.9249	16.8364
ABQ	25.39 ± 0.29	24.72 ± 0.04	25.92 ± 0.23	9.6779	17.3563	29.2905
BSS	36.91 ± 3.57	50.32 ± 0.10	57.47 ± 1.32	1.2842	1.2261	1.7796
RAH	**37.48** ± 1.60	**51.50** ± 0.04	**58.12** ± 2.20	1.2688	1.2444	1.7593

TABLE VI
GESTURE RECOGNITION PERFORMANCE ON CAMBRIDGE GESTURES

METHODS	MAP (%), $b = 4 \times 10^3, m = 10^4$	SEARCH TIME (S)
GLH	95.00 ± 0.30	0.2850
BSS	93.17 ± 0.93	0.1273
RAH	95.22 ± 1.27	0.0345

state-of-the-art vector hashing methods. Furthermore, different numbers of hash bits are used, our RAH method perform best consistently. Note that we here only adopt the features of 162 dimensions, meaning that the search time cost is similar for RAH and BSS. In practice, however, as the feature dimension rises, RAH's speedup will become much more important (see the speedup in Fig. 4(b) and the analysis in Table I). Moreover, the speed of RAH is more than 5× faster than the vector hashing approaches in this massive task.

C. Gesture Recognition

Action recognition has been widely studied in the literature. For instance, [61] proposed a spatio-temporal deformable ConvNet module with an attention mechanism to enhance the learning capacity for video action recognition. Here, in order to make the envaluation of the subspace search more simply and clearly, we choose the gesture recognition task which involves discrimination based on a number of similar successive images. Therefore, much like in fact recognition, each set of consecutive images can be represented by a subspace, while the recognition can be completed based on the nearest subspace search.

Dataset: In this experiment, we employ the Cambridge Gestures database [62], containing 900 videos of hand gestures [Fig. 3(c)]. There are three kinds of hand shapes—flat, spread, and v-shape — and three kinds of motions—left, right and contraction. Therefore, the combination of hand shapes and motions yield nine classes of gestures. Each of these classes contains 100 samples, which are evenly divided into five sets of 20 samples each. We use one set for testing and the remaining four sets as the database. For each video sample, the middle 32 frames are selected to represent the video; each of these frames is transformed into gray-scale and rescaled to 20 × 20.

Experiment Setup: We follow the same strategy used in [1]. Namely, we first unfold the 20 × 20 × 32 data cube along any of the three axes (x, y, t) to form three subspaces respectively. Next, three binary signatures with the same length are generated for the corresponding three subspaces. Finally, we concatenate the binary codes to form a single code for each data cube for the database and the testing set.

Results: Table VI reports the precision performance of the top-1 results and the search time. In this experiment, RAH performs similarly to or even better than the state-of-the-art subspace hashing methods GLH and BSS. Moreover, we can see that RAH significantly boosts the online search process, compared to BSS and GLH.

VI. CONCLUSION

In this paper, we mainly focused on the fast nearest neighbor search over large-scale subspace data, and further proposed a new type of locality-sensitive hashing approach, named random angular hashing (RAH), for linear subspaces of arbitrary dimension. RAH encodes the subspace by means of both random subspace hashing and binary quantization, and it can theoretically provide fast computation and strong locality sensitivity. The experiments on three popular tasks indicate that the RAH approach we proposed, through obtaining state-of-the-art performance, constitutes a satisfying method to the nearest subspace search problem.

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